THE UNIVERSITY OF HONG KONG

DEPARTMENT OF MATHEMATICS

MATH2012: Fundamental Concepts of Mathematics

May 20, 2019 2:30p.m. – 5:00p.m.

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

Answer ALL EIGHT questions

Note: You should always give precise and adequate explanations to support your conclusions. Clarity of presentation of your argument counts. So **think carefully before you write.**

1. (8 points) In the following, X and Y are sets while P is a statement. For each part, fill in the blank in the **simplest** way so that the sentence becomes meaningful and is always true. No explanation is required.

   (a) $(\mathbb{Z} \times$ _____ $) \subseteq (\mathbb{N} \times \mathbb{Z})$

   (b) $\forall\, a, b \in \{4, 6,$ _____ $\}$ with $a \neq b$, $\exists\, c \in \mathbb{Z}$ such that $ab + 1 = c^2$

   (c) $(\forall\, a \in X,\ P(a)) \wedge (\forall\, a \in Y,\ P(a)) \Leftrightarrow (\forall\, a \in (X$ _____ $Y),\ P(a))$

   (d) $f([1,2]) =$ _____ where $f : \mathbb{R} \to \mathbb{R}$ satisfies $f(x^3) = -x$ for any $x \in \mathbb{R}$

2. (20 points) Prove or disprove each of the following. State clearly whether it is true or false.

   (a) For any nonempty sets X and Y, we have $X \cup Y \neq X \cap Y$.

   (b) $\forall\, x \in \mathbb{R}$, $\exists\, y \in \mathbb{R}$ such that $y < x < y^2$.

   (c) For any set S with at least two elements, there is no bijective function from $S \times S$ to S.

   (d) Let $(G, *)$ be an abelian group. Then $(a * b)^{-1} = a^{-1} * b^{-1}$ for any $a, b \in G$.

   (e) There exists a binary operation $*$ on $\mathbb{Z}$ such that $(\mathbb{Z}, *)$ is a group for which 2 is the identity of the group.

3. (12 points) Let X and Y be disjoint sets and let Z be any set. Without using Venn diagrams, prove that $(X \setminus Z) = (X \setminus (Y \cup Z))$.

4. (a) (6 points) Prove that there is no square number of the form $n^2 + 6$ for some positive integer n.

   (b) (6 points) Prove that there exists a square number of the form $n^2 + 2019$ for some positive integer n.

5. Let R be a relation on a nonempty set S.

(a) (8 points) If R is an equivalence relation, prove that

$$((aRb) \wedge (aRc)) \Rightarrow ((bRc) \wedge (cRb))$$

for any $a, b, c \in S$.

(b) (4 points) Suppose $((aRb) \wedge (aRc)) \Rightarrow ((bRc) \wedge (cRb))$ for any $a, b, c \in S$. Prove or disprove that R is an equivalence relation.

6. (a) (8 points) Let $f : X \to Y$ and $g : Y \to Z$ be functions where X, Y and Z are nonempty sets. For any subset S of Z, prove that

$$(g \circ f)^{-1}(S) = f^{-1}(g^{-1}(S)).$$

(b) (4 points) Define $f : \mathbb{R}^2 \to \mathbb{R}^2$ and $g : \mathbb{R}^2 \to \mathbb{R}$ by

$$f((x, y)) = (x^2 - xy + y^2, (x - y)^2) \quad \text{and} \quad g((x, y)) = x - y$$

for any $x, y \in \mathbb{R}$. Find $f^{-1}(g^{-1}(\{1\}))$.

7. (a) (4 points) Let $S = \{1, 2, 3, 4, 5\}$ and let $\mathcal{A} = \{\{1, 2\}, \{3, 4\}, \{5\}\}$ be a partition of S. Suppose this partition is induced by the equivalence relation R. Find the cardinality of R.

(b) (8 points) Let $\mathcal{B}$ be a partition of $\mathbb{R}$ such that each element of $\mathcal{B}$ is a nontrivial interval (i.e. a subset of the form $(a, b), [a, b], [a, b)$ or $(a, b]$ for some real numbers $a < b$). Prove that $\mathcal{B}$ is countable. (Hint: You can make use of the following fact that we have proven: for any real numbers a and b with $a < b$, there exists a rational number x satisfying $a < x < b$.)

8. (a) (4 points) Let p be a prime number and let n be an integer. Using mathematical induction, prove that $n^{k(p-1)+1} \equiv n \pmod{p}$ for any positive integer k. (Hint: You can use the following fact without proof: for any prime number p and any integer n, we have $n^p \equiv n \pmod{p}$.)

In parts (b) and (c), suppose p is a prime number of the form $3m+2$ for some integer m.

(b) (4 points) If a and b are integers satisfying $a^3 \equiv b^3 \pmod{p}$, prove that $a \equiv b \pmod{p}$.

(c) (4 points) Using (b) or otherwise, prove that for any integer c, there exists an integer x such that $x^3 \equiv c \pmod{p}$.

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